

RGB LED

DO

OUTPUT

LIGHT

What does it do?

An RGB LED puts red, green and blue lights inside one clear package.

Varying the brightness of the three colours mixes millions of possible colours.

A common-cathode LED has one shared negative leg and needs one resistor for each colour.

Uses



Wi-Fi router status

DETECT
Read connection status

DECIDE
Which status colour is needed?

DO
Mix the LED colour



Mood crystal

DETECT
Read the mood dial

DECIDE
What is the crystal's mood?

DO
Glow that colour



Clear lens
Contains three LEDs

Long leg
Common cathode

R
One 220 ohm resistor each

Code

● Raspberry Pi

```
from gpiozero import RGBLED
light = RGBLED(red=17, green=27, blue=22)
light.color = (1, 0, 1)
```

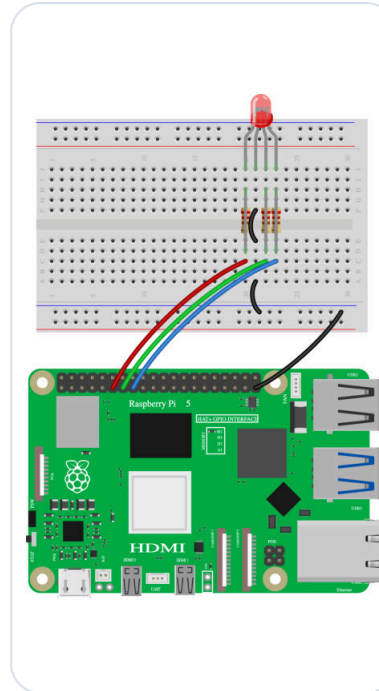
● Raspberry Pi Pico

```
from picozero import RGBLED
light = RGBLED(red=15, green=13, blue=10)
light.color = (255, 0, 255)
```

Pi colour values run from 0 to 1; picozero colour values run from 0 to 255. These examples make magenta.

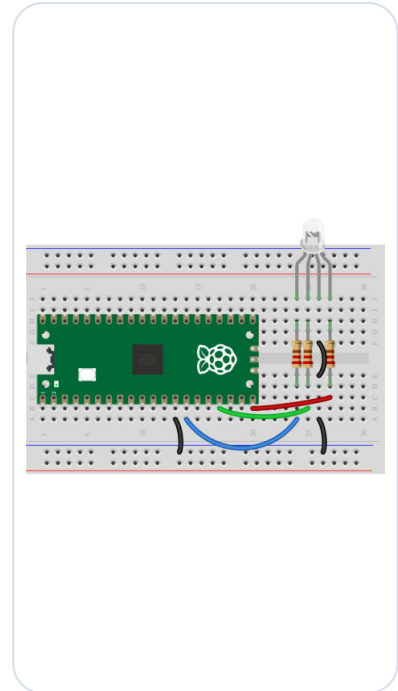
Wiring

RASPBERRY PI 5



R → **GPIO17**, G → **GPIO27**, B → **GPIO22**;
common → **GND**

PICO



R → **GP15**, G → **GP13**, B → **GP10**;
common → **GND**